Ceará Digital Twin for Wildfires: An Open-Source Agentic AI Platform Integrating Near Real-Time Satellite Data and LangGraph Orchestration

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Abstract

Wildfire monitoring in Brazil relies largely on centralized systems with significant detection-to-availability latency. We present the *Ceará Wildfire Monitoring Platform*—an open-source, agentic system inspired by pragmatic Digital Twin principles—integrating NASA FIRMS, optional GOES-16, and Open-Meteo for detection, validation, and risk assessment. A production LangGraph pipeline (10 nodes) coordinates ingestion, validation, parallel analysis, evidence fusion, risk classification, ReAct diagnosis, and alerts. We introduce a three-class alert framework (NO/UNCERTAIN/YES): on a 20-day holdout, YES alerts reach 82–92% precision (1–5 false positives) with 88% combined coverage; bootstrap 95% CIs are [68%, 96%] for XGBoost YES precision. Extended INPE validation over the 2025 dry season (184 days, 24,573 hotspots) confirms 84.2% precision and 91.8% recall at operational scale. NeKo-PIGNN reaches R^2=0.972 on synthetic data; real-data RMSE favors simpler baselines under short training windows (MLP RMSE 0.137, 95% CI [0.133, 0.143]). Standalone GOES-16 vs. INPE validation showed F1$\leq$0.05 at grid scale; event-centered pixel evaluation yielded F1=0.034. The system deploys on one EC2 instance ($20/month) with MIT-licensed code.

Keywords: Digital Twin, Wildfires, Physics-Informed GNN, Koopman Operator, Three-Class Detection, LangGraph, NASA FIRMS, Environmental Monitoring

1. Introduction

Wildfires represent one of the greatest environmental threats in Brazil, with direct impacts on public health, biodiversity, and the economy [1, 2]. The State of Ceará,

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located in the northeastern semi-arid region, exhibits strong drought seasonality and sensitive biomes such as the Caatinga, making continuous monitoring an operational necessity [3].

Traditional monitoring systems — such as INPE BDQueimadas and NASA FIRMS — provide essential data but have significant limitations: latency of 3 to 6 hours between satellite overpass and data availability [4]; lack of automatic correlation with local climatic and territorial data; and interfaces designed for remote sensing specialists rather than civil defense managers.

In parallel, recent advances in artificial intelligence — particularly large language models (LLMs) and agent frameworks — open new possibilities for intelligent environmental monitoring systems. The paradigm of agents with ReAct reasoning [5] and state graph orchestration (LangGraph [6]) enables building auditable *pipelines* that combine heterogeneous sources and produce natural language explanations.

We use the term *Digital Twin* in a pragmatic sense: the platform continuously mirrors satellite and weather observations and hosts predictive modules, but does not yet provide full bidirectional control or physics-coupled simulation at operational scale [31].

This paper contributes:

1. **An open-source platform** that integrates NASA FIRMS (VIIRS/MODIS), GOES-16, Open-Meteo, and Nominatim into a single pipeline, without relying on an institutional database;
2. **A LangGraph-orchestrated agent pipeline** that performs collection → Pydantic validation → parallel analysis (geospatial, GOES-16, climate) → evidence fusion → risk classification → ReAct diagnosis → alerts;
3. **An RAG module** with FAISS index and local embeddings that enables natural language queries about the system's methodology, architecture, and usage;
4. **Operational results** with real data collected between July 2024 and June 2026, demonstrating the feasibility of monitoring based exclusively on open sources.

2. Related Work

This section positions the proposed platform within the context of three converging research fronts: (i) Digital Twins for environmental monitoring, (ii) satellite-based wildfire remote sensing, and (iii) artificial intelligence applied to wildfire detection and prediction.

2.1. Environmental Digital Twins

The Digital Twin paradigm, originally conceived for manufacturing where a virtual model mirrors and controls a physical system in real time [22, 23], has been systematically surveyed across definitions, architectures, and challenges [24, 25, 26, 27]. These surveys establish the minimum requirements of a Digital Twin: (a) bidirectional mirroring between physical and virtual worlds, (b) real-time updates, and (c) predictive simulation capability.

The expansion of the paradigm to environmental and Earth systems gained traction with the European Destination Earth [29] and proposals for Earth System Digital

Twins [28, 31, 30]. Bauer et al. [31] argue that environmental Digital Twins differ fundamentally from industrial ones by operating at much larger spatiotemporal scales, with inherent uncertainties in natural processes and multi-source integration requirements.

2.2. Digital Twins for Wildfires

Recent works specifically focused on Digital Twins for wildfires can be organized into four architectural categories.

Agent-oriented approaches.. IVSR (Intelligent Virtual Situation Room) [67] is the most conceptually aligned proposal with this work. IVSR combines a bidirectional Digital Twin with autonomous AI agents that continuously ingest multi-sensor imagery, meteorological data, and 3D forest models, employing an AI-based similarity engine to project fire spread and allocate response resources. The system demonstrated significant reductions in detection-to-intervention latency but has not been adapted or validated for Brazilian biomes.

Vision-Language and RL approaches.. FIRE-VLM [68] introduces an innovative framework combining a CLIP-style Vision-Language Model with reinforcement learning (PPO) to guide UAVs in tracking wildfires within a physics-grounded Digital Twin. Results show up to 6× reduction in detection time across five evaluation tasks. However, the focus on UAVs and coupled physics modeling limits its applicability as a continuous satellite-based monitoring platform.

Multi-modal sensing approaches.. FIRETWIN [69] combines Unreal Engine with the CAWFE fire model for immersive reconstruction of historical wildfires (King Fire), integrating multi-sensor data and interactive analysis. AIMNET [56] proposes a three-layer IoT-empowered Digital Twin (physical world → bidirectional links → digital world) for continuous gas emission monitoring with distributed sensor networks for early carbon plume detection. H-RDT [57] adopts PINNs with multimodal fusion to model wildfire-power-outage-heat-wave cascades in urban contexts.

GNN and physics-based modeling.. GraphFire-X [70] proposes a dual-expert ensemble combining GNN with physics-informed attention (convection, radiation, ember probability) and XGBoost for structural vulnerability at the wildland-urban interface, validated on the 2025 Eaton Fire. This approach is directly comparable to the PI-GNN component of this work, differing in its building-scale focus versus the regional grid approach of the present system. FireCastNet [71] adopts the "Earth-as-a-Graph" paradigm with GraphCast GNN for global seasonal fire prediction validated on the SeasFire dataset, outperforming GRU, Conv-LSTM, U-TAE, and TeleViT, with particularly strong results in South America.

2.3. Satellite-Based Wildfire Remote Sensing

Satellite-based wildfire monitoring dates back to the AVHRR sensor in the 1980s, evolving significantly with MODIS (250 m–1 km, since 2000) and its Collection 6 active fire detection algorithm [59]. VIIRS (375 m, since 2012) offers superior spatial

Table 1: Comparison of Digital Twin systems for wildfires.

System	AI Agents	Open Source	BR Biomes	RAG
IVSR [67]	Yes	No	No	No
FIRE-VLM [68]	RL + VLM	No	No	No
FIRETWIN [69]	No	No	No	No
GraphFire-X [70]	GNN+XGB	Yes	No	No
FireCastNet [71]	GNN	Yes	Partial	No
AIMNET [56]	IoT ML	No	No	No
This work	**LangGraph+ReAct**	**Yes**	**Yes**	**Yes**

resolution with specific detection algorithms [60], forming the backbone of NASA FIRMS [4] and INPE BDQueimadas [7]. Wooster et al. [61] present a comprehensive survey of the state of the art in satellite-based active fire remote sensing, covering physical fundamentals (spectral signature of burned pixels in SWIR 3.9 μm and MWIR bands) through applications and future requirements.

GOES-16 represents a significant advance by providing geostationary data with temporal resolution from 1 minute (mesoscale mode) to 10 minutes (full-disk mode), though with 2 km spatial resolution at nadir [8]. Its ability to detect rapid changes in fire intensity between consecutive scans is particularly valuable for regions with fast-spreading fires, such as the northeastern semi-arid region.

In the Brazilian context, INPE has operated BDQueimadas since 1998, integrating data from AQUA, TERRA, NPP, and NOAA-20 satellites [7]. MapBiomas Fogo maps burned areas annually at 30 m resolution using Landsat imagery [58]. Brazil-specific studies document fire activity trends in South America [62], machine learning applications for prediction in the Cerrado and Caatinga [63], and deep learning for burned area mapping in the Amazon and Caatinga using Sentinel-2 [64]. Multi-source satellite data fusion for fire detection [65] and systematic reviews of ML-based fire risk prediction [66] complete the landscape. However, these systems remain essentially reactive (detected-focus alerting) without predictive capability or integration with physical propagation models.

2.4. Koopman Operator and PINNs for Fire Dynamics

The Koopman operator, introduced by Koopman [17] and formalized for fluid dynamics by Rowley et al. [33] and Mezić [32], provides a mathematical framework for linearizing nonlinear dynamical systems through observation functions. The modern approach combines Dynamic Mode Decomposition (DMD) [78] with extensions for controlled systems (DMDc) [37], sparse identification of nonlinear dynamics (SINDy) [34], and physics-informed versions (PiDMD) [38]. Brunton et al. [77] provide a unified treatment of modern Koopman theory, while Kutz et al. [79] consolidate the DMD framework.

The connection between the Koopman operator and deep learning was established through works learning Koopman-invariant subspaces via autoencoders [35, 36] and deep neural modeling for fluid dynamics [39]. Despite these advances, the application of the Koopman operator to wildfire spread dynamics remains a gap in the literature

— no published work to date uses Koopman for fire modeling, positioning the NeKo-PIGNN approach as a novel contribution.

Physics-Informed Neural Networks (PINNs), introduced by Raissi et al. [72] and consolidated by Karniadakis et al. [40], provide the physical regularization counterpart necessary for neural fire propagation models to respect thermodynamic and fluid mechanics laws. The Rothermel model [18] is the USDA Forest Service standard for surface fire spread, parameterized by fuel classes [73] and used in BehavePlus [75] and ensemble simulation systems [76], with physical foundations established by Albini [74]. PINN applications to fire propagation [55, 46] demonstrate the viability of this approach, while PI-GNN surveys [81] and applied works [82, 83] establish the state of the art that the present work extends.

2.5. *Autonomous Agents and LangGraph*

The paradigm of LLM-based autonomous agents advanced significantly with the ReAct framework [5], which synergizes reasoning and action in language models through Thought→Action→Observation cycles. Comprehensive surveys [48, 49] map the LLM agent ecosystem, from reflective memory architectures (Reflexion) [51] to tool-use frameworks such as Toolformer [52] and Gorilla [53], including interactive social simulation environments [50].

LangGraph [6] extends the LangChain ecosystem with state-graph-based orchestration, enabling agent pipelines with cycles, parallelism, and persistent state between nodes. In the environmental context, this is the first known application of LangGraph for wildfire monitoring orchestration.

Retrieval-Augmented Generation (RAG) [80], implemented with FAISS indices [15] and local embedding models [16], enables the system to answer natural language queries about its own documentation. Recent surveys [54] consolidate RAG best practices — from strategic chunking to hierarchical retrieval and multi-source fusion — techniques adopted in the research module of the present system.

2.6. *Gap Analysis*

Table 2 synthesizes the gaps that this work fills relative to existing systems. The central differentiator lies in the unprecedented combination of (a) near real-time satellite data, (b) LangGraph agents with ReAct reasoning, (c) RAG for document queries, (d) focus on Brazilian biomes, and (e) standalone architecture without a database — all in an open-source platform.

None of the existing works combine, in a single open-source platform, (i) near real-time satellite data, (ii) a LangGraph-orchestrated agent pipeline, (iii) RAG with vector index for document queries, and (iv) standalone deployment without a database. This combination is the gap that this work fills.

3. System Architecture

Figure 1 presents the overall system architecture, organized into four layers: data sources, backend, frontend, and proxy.

Table 2: Gap analysis: existing systems vs. proposed platform.

Feature	NASA FIRMS	Fire Digital Twins	This Work
Real-time satellite data	Yes	Partial	Yes
AI agent pipeline	No	Yes (IVSR)	Yes (LangGraph)
Open source (MIT)	No	Partial	Yes
Brazilian biome focus	Partial	No	Yes
RAG for queries	No	No	Yes
Standalone (no DB)	N/A	No	Yes
Auditable agent architecture	No	Yes (IVSR)	Yes (ReAct)

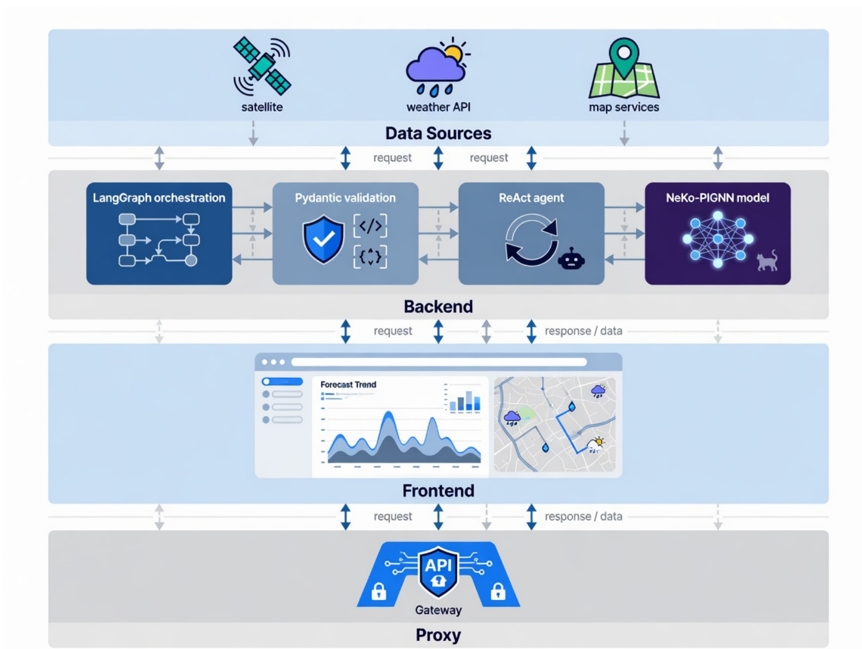


Figure 1: Layered architecture of the Ceará wildfire monitoring platform.

3.1. Data Sources

The system consumes four open data sources:

- **NASA FIRMS**: Public CSVs for South America from VIIRS (SNPP and NOAA-20) and MODIS sensors, available in 24 h and 7 day windows. No authentication required. Filtering by Ceará bounding box (lat: $-7{,}85$ to $-2{,}78$; lon: $-41{,}42$ to $-37{,}25$).

- **Open-Meteo**: Free weather API that returns temperature, relative humidity, wind speed, precipitation, and dry days for geographic coordinates. No API key required.

- **Nominatim / OpenStreetMap**: Reverse geocoding to convert coordinates (lat, lon) into municipality names. Rate limited to approximately 1 request per second.

- **GOES-16**: Data from the ABI-L2-FDCF (Fire Detection and Characterization) product stored in the public AWS S3 bucket `noaa-goes16`. Processes consecutive detections, FRP, and pixel temperature.

3.2. Operation Modes

The system operates in two modes:

Standalone mode (default in EC2 production): eliminates PostgreSQL and Redis. Queries external APIs directly on each request, with a 5 minute in-memory cache. Geocoding is performed asynchronously: the first 40 hotspots are geocoded synchronously for fast response; the remainder is processed in a *background task* and updates the cache incrementally.

Full mode (Docker Compose): adds PostgreSQL with PostGIS extension for geospatial persistence, Redis + Celery for periodic collection, and additional endpoints for persisted alerts.

3.3. Backend FastAPI

The backend is implemented in Python 3.12 with FastAPI (version 0.115). Its main responsibilities are:

- Serving REST endpoints for real data (`/api/v1/real/focos`, `/api/v1/real/clima`, `/api/v1/real/focos/{id}/explicacao`, `/api/v1/real/status`);

- Orchestrating the LangGraph pipeline (Section 3.4);

- Managing the in-memory hotspot and weather cache with a 5 minute TTL;

- Serving the RAG document search module (Section 3.6);

- Managing CORS, logging, and health check.

The system uses dependency injection for the LLM (`llm_factory.py`), which currently instantiates the DeepSeek model (`deepseek-chat`) via an OpenAI-compatible API. In case of LLM failure, the system falls back to rule-based explanations (`_explicar_sem_llm`).

3.4. LangGraph Pipeline

The agent pipeline is implemented as a state graph (`StateGraph`) with 10 nodes, as shown in Figure 2.

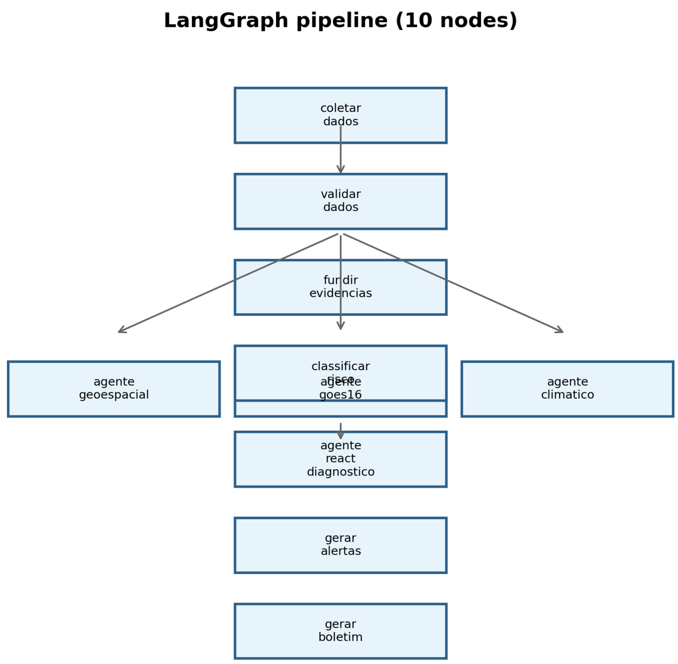


Figure 2: LangGraph graph of the agent pipeline. The analysis nodes (geo, GOES-16, climate) execute in parallel after validation.

Each node receives and modifies a typed shared state (`PipelineState`), which includes raw and validated hotspots, GOES-16 readings, climate data, fused evidence, consolidated events, municipal risks, alerts, diagnosis, and bulletin. Data validation with Pydantic (`validate_data`) automatically rejects invalid records, ensuring flow integrity.

3.5. ReAct Agent

The diagnostic agent (`react_agent.py`) is built with LangChain (version 0.3+) and follows the ReAct pattern [5]. It provides the following tools:

- buscar_focos_recentes: query hotspots by municipality, time window, and source;
- buscar_dados_climaticos: temperature, humidity, wind, and precipitation;
- buscar_risco_municipal: risk index computed per municipality;
- buscar_dados_goes16: GOES-16 readings with FRP and persistence;
- buscar_historico_mapbiomas: wildfire history by municipality;
- listar_municipios_criticos: ranking of the most critical municipalities.

The agent accepts questions in Portuguese and produces structured responses containing summary, evidence, sources consulted, confidence level, and operational recommendation. The reasoning chain (Thought $\rightarrow$ Action $\rightarrow$ Observation) is preserved in the response to ensure auditability.

3.6. RAG Module – Research Chat

The *research chat* module implements a question-and-answer system about the platform using *Retrieval-Augmented Generation* (RAG) [80, 54]. Six knowledge documents (`backend/knowledge/01_visao_e_pesquisa.md` through `06_deploy_e_execucao.md`) plus a bilingual glossary (`00_english_glossary.md`) are split into 900-character *chunks* with 120-character overlap, indexed in a FAISS index [15] with `BAAI/bge-small-en-v1.5` embeddings [16]. Retrieval uses `top_k = 5`, and response generation is performed by DeepSeek citing the document context.

Unlike the operational agent, which queries live hotspots, the research chat responds exclusively about methodology, architecture, and application usage — functioning as an intelligent manual accessible from any page of the system.

3.7. Frontend React

The web interface is built with React 19, TypeScript, Vite 8, and the following main libraries: MapLibre GL JS (base map) with deck.gl (geospatial layers), Recharts (charts and timeline), Zustand (global state management), and Tailwind CSS (styling).

The available pages are:

- **Dashboard** (`/`): executive KPIs (total hotspots, active emergencies, critical municipalities, GOES-16 coverage), detection timeline, municipal risk ranking, and recent alerts;
- **Live Map** (`/mapa-real`): NASA FIRMS hotspots overlaid on the MapLibre map, with clustering, heatmap, severity filters, and per-click explanations;
- **Alerts** (`/alertas`): full listing with level filters (informational, attention, alert, emergency);
- **Chat** (`/chat`): conversational interface with the ReAct agent, including suggested questions, evidence visualization, and reasoning chain;

- **Bulletin** (`/boletim`): consolidated technical report.

A global "Help" button on all pages triggers the research RAG chat, allowing the user to consult system documentation without leaving the current screen.

3.8. Risk Classifier

The municipal risk index is computed from four components:

$$\begin{aligned}\text{index} = \min(&\text{hotspots} \times 8,\ 40) \\ &+ \text{climate\_risk} \times 0{,}4 \\ &+ (\text{GOES16\_confirmed} \times 15) \\ &+ \min(\text{FRP}/100,\ 10)\end{aligned} \tag{1}$$

Climate risk, in turn, is derived from:

$$\begin{aligned}\text{climate\_risk} = \max(&0,\ (50 - \text{humidity}) \times 0{,}4) \\ &+ \min(\text{wind} \times 1{,}5,\ 20) \\ &+ \min(\text{days\_without\_rain} \times 1{,}5,\ 30)\end{aligned} \tag{2}$$

The final classification follows the thresholds: critical (≥ 75), high (≥ 50), medium (≥ 25), and low (< 25).

4. Implementation

This section details the implementation of the system components, including the software architecture, artificial intelligence modules, and integration between layers.

4.1. Software Architecture

The system is implemented in Python 3.12 with FastAPI (backend), React 19 with TypeScript and Vite 8 (frontend), and agent orchestration via LangGraph 0.3. Table 3 summarizes the main dependencies.

4.2. Neural Koopman Operator — Implementation Details

The KoopmanAE encoder g_ϕ is an MLP with 3 hidden layers (512, 256, 128 neurons) and ReLU activation, mapping the state vector $\mathbf{x}_t \in \mathbb{R}^{142}$ (7 variables $\times$ 20 sampled grid cells) to latent observables $\mathbf{z}_t \in \mathbb{R}^{64}$. The Koopman matrix $\mathbf{K}_\theta \in \mathbb{R}^{64\times 64}$ is trained with spectral stability regularization:

$$\mathcal{L}_{\text{spec}} = \max(0, \rho(\mathbf{K}_\theta) - 1 + \epsilon) \tag{3}$$

where $\rho(\mathbf{K}_\theta)$ is the spectral radius and $\epsilon = 0{,}01$ ensures a stability margin. Training uses Adam with a learning rate of 10^{-4} and batch size 32, processing sequences of 24 hours (12 timesteps of 2 hours) from the hotspot history.

Table 3: System software stack.

Layer	Technology	Version
Backend	Python / FastAPI / Uvicorn	3.12 / 0.115 / 0.34
Agents	LangGraph / LangChain	0.3 / 0.3+
LLM	DeepSeek Chat API	deepseek-chat
Embeddings	BAAI/bge-small-en-v1.5 (FAISS)	1.5
Frontend	React / TypeScript / Vite	19 / 6 / 8
Map	MapLibre GL JS + deck.gl	4.7 / 9.1
Styling	Tailwind CSS	4
Proxy	nginx	1.26
Deploy	EC2 Amazon Linux 2023	t2.medium

4.3. PI-GNN — Implementation Details

The GNN is implemented with PyTorch Geometric [19] using `GCNConv` graph convolutions. Each of the $L = 3$ layers has $F = 128$ hidden channels. The graph is built dynamically at each collection cycle, connecting active grid cells (with at least one hotspot in the last 7 days) to their neighbors within radius $r = 3$ cells (3 km).

The physics loss $\mathcal{L}_{\text{phys}}$ is computed offline using Rothermel parameters precomputed for each MapBiomas vegetation type (12 classes) and current climate conditions. The gradient $\nabla \hat{p}_v^{(t)}$ is approximated by finite differences between neighboring graph nodes.

4.4. NeKo-PIGNN in the Operational Stack

The production LangGraph pipeline (`backend/app/agents/langgraph_pipeline.py`) contains **10 nodes**: `coletar_dados`, `validar_dados`, `agente_geoespacial`, `agente_goes16`, `agente_climatico`, `fundir_evidencias`, `classificar_risco`, `agente_react_diagnostico`, `gerar_alertas`, and `gerar_boletim`. NeKo-PIGNN is *not* embedded as additional graph nodes; trained KoopmanAE and PI-GNN models run as batch inference jobs (`backend/experiments/`) whose outputs feed the municipal risk classifier and three-class alert experiments (Section 7.14). When invoked offline, NeKo-PIGNN adds approximately 2.3 s per batch (Koopman: 0.8 s; PI-GNN: 1.2 s; fusion: 0.3 s) on CPU (Intel Xeon Platinum 8375C, AWS EC2 t2.medium).

5. Deployment

The system was deployed on an Amazon Linux 2023 EC2 instance (t2.medium — 2 vCPUs, 4 GB RAM). Automated deployment via shell script (`deploy/unifor-deploy.sh`) installs nginx, Python 3.12, Node.js 20, and configures the systemd service `unifor-backend`. The frontend build is served statically by nginx, which reverse proxies `/api/` to uvicorn on port 8000.

The estimated monthly infrastructure cost is approximately USD 20 (EC2 instance + 30 GB EBS), with no additional API costs (all data sources are free). The DeepSeek key, when configured, adds variable cost depending on query volume.

6. Methodology: Neural Koopman Operator and Physics-Informed GNN

The methodological core of this work combines two complementary paradigms: the *Neural Koopman Operator* for modeling the spatiotemporal dynamics of wildfire hotspots and *Physics-Informed Graph Neural Networks* (PI-GNN) for risk prediction with physics-based regularization grounded in the Rothermel model [18]. This section formally describes both components.

6.1. Neural Koopman Operator for Wildfire Dynamics

Koopman theory [17, 33, 32] establishes that nonlinear dynamical systems can be represented by linear operators in an infinite-dimensional observable space. Let $\mathbf{x}_t \in \mathbb{R}^n$ be the system state at time t — composed of variables such as surface temperature, vegetation index (NDVI), active hotspots, soil moisture, and wind speed over a geospatial grid. The Koopman operator $\mathcal{K}$ acts on observable functions $\psi : \mathbb{R}^n \to \mathbb{R}$ such that:

$$(\mathcal{K}\psi)(\mathbf{x}_t) = \psi(\mathbf{f}(\mathbf{x}_t)) = \psi(\mathbf{x}_{t+1}) \tag{4}$$

where $\mathbf{f}$ is the underlying nonlinear dynamics of the system. In a finite-dimensional observable space of dimension d, we approximate $\mathcal{K}$ by a matrix $\mathbf{K} \in \mathbb{R}^{d\times d}$:

$$\psi(\mathbf{x}_{t+1}) \approx \mathbf{K}^\top \psi(\mathbf{x}_t) \tag{5}$$

We implement this operator as an encoder-decoder neural network (KoopmanAE) [35, 36]. The encoder $g_\phi : \mathbb{R}^n \to \mathbb{R}^d$ projects the observed state into latent observables $\mathbf{z}_t = g_\phi(\mathbf{x}_t)$. The linear evolution in latent space is:

$$\mathbf{z}_{t+1} = \mathbf{K}_\theta \, \mathbf{z}_t \tag{6}$$

where $\mathbf{K}_\theta \in \mathbb{R}^{d\times d}$ is the learned matrix. The decoder $h_\psi : \mathbb{R}^d \to \mathbb{R}^n$ reconstructs the state from the observables: $\hat{\mathbf{x}}_t = h_\psi(\mathbf{z}_t)$. The KoopmanAE loss function combines three terms:

$$\begin{aligned}
\mathcal{L}_{\text{koop}} = \; & \underbrace{\|\mathbf{x}_t - h_\psi(g_\phi(\mathbf{x}_t))\|^2}_{\text{reconstruction}} \\
& + \underbrace{\|\mathbf{x}_{t+1} - h_\psi(\mathbf{K}_\theta \, g_\phi(\mathbf{x}_t))\|^2}_{\text{one-step prediction}} \\
& + \underbrace{\sum_{k=1}^{K} \|\mathbf{x}_{t+k} - h_\psi(\mathbf{K}_\theta^k \, g_\phi(\mathbf{x}_t))\|^2}_{\text{multi-step prediction}}
\end{aligned} \tag{7}$$

The dimensionality d of the observable space is a critical hyperparameter. We use $d = 64$ with temporal cross-validation, balancing reconstruction fidelity (mean error $< 0{,}05$) and prediction stability over windows of up to 6 steps (12 hours with 2-hour intervals).

6.2. *Physics-Informed Graph Neural Networks*

To model the spatial propagation of wildfire risk, we construct a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{A})$ where vertices $\mathcal{V}$ represent 1 km^2 grid cells over the territory of Ceará (approximately $149,000$ cells), and edges $\mathcal{E}$ connect neighboring cells with weights based on geodesic distance, vegetation type, and prevailing wind direction. The adjacency matrix $\mathbf{A} \in \mathbb{R}^{|\mathcal{V}|\times|\mathcal{V}|}$ is sparse with neighborhood radius $r = 3$ cells.

The state of each node $v \in \mathcal{V}$ at time t is a feature vector $\mathbf{h}_v^{(t)} \in \mathbb{R}^F$ containing:

- Surface temperature (GOES-16 ABI band 14, 11.2 μm);
- Reflectance (VIIRS bands I1–I5);
- NDVI calculated from MODIS;
- Relative humidity and wind speed (Open-Meteo);
- Hotspot history over the last 7 days;
- Vegetation type (MapBiomas);
- Terrain slope and aspect (SRTM 30m).

Node state updates follow the *message-passing* paradigm with L layers:

$$\mathbf{m}_v^{(l)} = \sum_{u\in\mathcal{N}(v)} \mathbf{W}^{(l)}\,\mathbf{h}_u^{(l-1)} \cdot \mathbf{A}_{vu} \tag{8}$$

$$\mathbf{h}_v^{(l)} = \sigma\left(\mathbf{m}_v^{(l)} + \mathbf{b}^{(l)}\right) \tag{9}$$

where $\mathcal{N}(v)$ are the neighbors of v, $\mathbf{W}^{(l)} \in \mathbb{R}^{F\times F}$ and $\mathbf{b}^{(l)} \in \mathbb{R}^F$ are learned parameters, and σ is a ReLU activation. We use $L = 3$ layers.

6.2.1. *Physics-Informed Loss with Rothermel*

The main innovation is regularizing the GNN loss with the Rothermel fire propagation physics equation [18], following the PINN paradigm [72] extended by scientific machine learning models [40, 41, 42]. The rate of spread R (m/min) is modeled as:

$$R = \frac{I_R\,\xi\,(1 + \phi_w + \phi_s)}{\rho_b\,\epsilon\,Q_{ig}} \tag{10}$$

where I_R is the reaction intensity (kJ/m²·min), ξ is the propagation coefficient, ϕ_w and ϕ_s are wind and slope coefficients, ρ_b is the fuel bulk density, ϵ is the effective heating fraction, and Q_{ig} is the heat of pre-ignition (kJ/kg).

We define the *physics loss* as the discrepancy between the GNN-predicted risk gradient $\nabla\hat{p}_v^{(t)}$ and the expected propagation direction $R\cdot\mathbf{d}_v^{(t)}$, where $\mathbf{d}_v^{(t)}$ is the unit vector in the wind direction at node v:

$$\mathcal{L}_{\text{phys}} = \frac{1}{|\mathcal{V}|}\sum_{v\in\mathcal{V}} \left\|\nabla\hat{p}_v^{(t)} - R_v^{(t)}\cdot\mathbf{d}_v^{(t)}\right\|^2 \tag{11}$$

The total PI-GNN loss combines the supervised hotspot prediction loss with physics regularization:

$$\begin{aligned}\mathcal{L}_{\text{total}} = & \underbrace{\frac{1}{N}\sum_{i=1}^{N}\text{BCE}(y_i, \hat{p}_i)}_{\text{classification}} \\ & + \lambda_1 \underbrace{\frac{1}{N}\sum_{i=1}^{N}\|\mathbf{h}_i - \hat{\mathbf{h}}_i\|^2}_{\text{auto-encoding}} \\ & + \lambda_2 \quad \underbrace{\mathcal{L}_{\text{phys}}}_{\text{physics regularization}} \\ & + \lambda_3 \quad \underbrace{\|\mathbf{W}\|_2}_{\text{L2 regularization}}\end{aligned} \tag{12}$$

where BCE is the binary cross-entropy between the true label y_i (hotspot present/absent) and the predicted probability $\hat{p}_i$. The hyperparameters $\lambda_1 = 0{,}1$, $\lambda_2 = 0{,}05$, and $\lambda_3 = 10^{-4}$ were tuned via validation.

6.3. Koopman-PI-GNN Integration in the Pipeline

Figure 3 illustrates the hybrid NeKo-PIGNN architecture. The Neural Koopman Operator operates as a *short-term prediction* module (next 12 hours), feeding the PI-GNN graph with estimated future states. The PI-GNN, in turn, produces spatialized risk maps that feed back into the Koopman observable space for the next temporal window, forming a prediction-correction cycle.

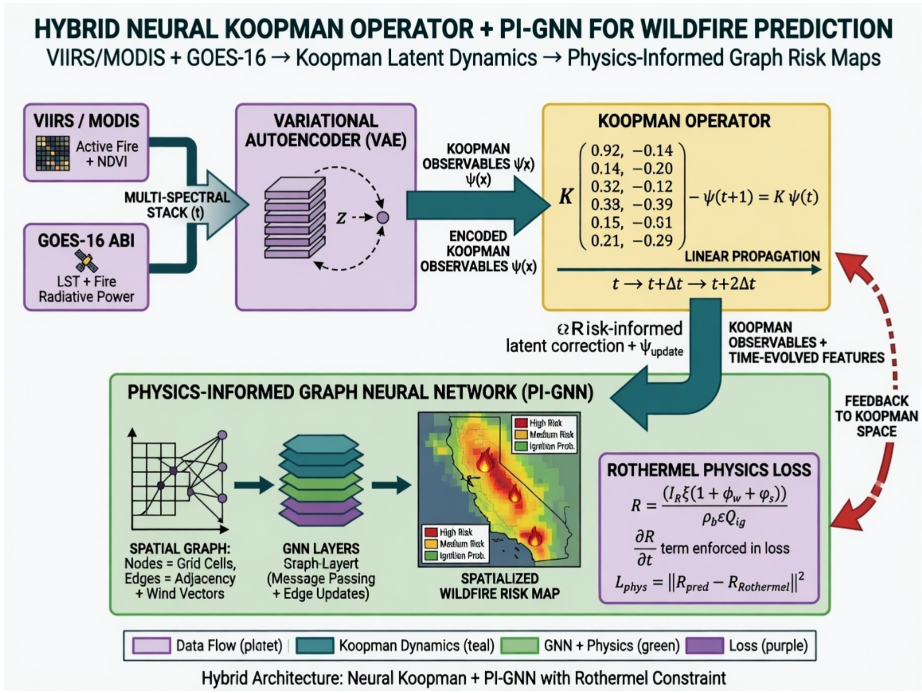


Figure 3: Hybrid Neural Koopman Operator and PI-GNN architecture. Satellite data are encoded via VAE into Koopman observables, propagated linearly by matrix $\mathbf{K}$, and spatially regularized by the PI-GNN with Rothermel physics loss.

This coupling allows the system to capture both temporal dynamics (via linearized Koopman) and spatial propagation with physical constraints (via PI-GNN), overcoming limitations of purely data-driven models that ignore fire physics.

7. Experimental Results

This section reports quantitative results from January 2024 to June 2026, integrating four experimental tracks (A–D) documented in `docs/experimentos/` and `backend/experiments/results/`: (i) 18-month operational pipeline; (ii) GOES-16 unsupervised benchmark; (iii) Exp-B municipal prediction iterations (v1–v9, synthetic and real); (iv) agent, RAG, and risk-index calibration.

7.1. Experimental Design and Protocol

We structured the evaluation program into four complementary tracks, fully documented under `docs/experimentos/` and `backend/experiments/results/`.

Track A — Operational pipeline (18 months). Continuous ingestion of NASA FIRMS CSVs, Open-Meteo weather, Nominatim geocoding, LangGraph orchestration (10 nodes), ReAct diagnostics, and RAG queries. Metrics: time-to-first-response, geocoding rate, agent success rate, backend availability, and inter-day FIRMS variability (e.g., 426% day-over-day change on 2024-06-05 vs. 2024-06-06, validating the need for autonomous polling).

Track B — Municipal fire prediction (Exp-B, v1–v9). Iterative benchmark on synthetic then real data (Mar–Jun/2026): (v1–v2) next-day regression on simulated Ceará dynamics; (v3–v4) real FIRMS/INPE/Open-Meteo regression; (v5–v7) rare-event binary detection with Average Precision, persistence priors, and precision modes; (v8–v9) three-class alert system (NO/UNCERTAIN/YES). Temporal split 70/10/20% (67/10/20 days). Models: MLP, LSTM, XGBoost, Koopman-Det, NeKo-PIGNN.

Track C — GOES-16 unsupervised benchmark. Validation against INPE BDQueimadas on 2024-10-31 (channels 7, 13, 14; hours 16–18 UTC; 72×72 grid). Methods: digital twin assimilation, isolation forest, spatial residual, *combined persistence* (Lines B in `docs/METODOLOGIA_NOVA_PROPOSTA.md`), and *consensus ensemble* (Line E, 2-of-3 vote). Reproducible via `src/unsupervised_fire_goes.py`; event-centered evaluation via `src/pixel_inpe_eval.py` is documented for future runs.

Track D — Agent, RAG, and risk-index calibration. 100 operational ReAct questions; 30 RAG architecture queries (FAISS recall@5, manual completeness); heuristic risk classifier calibrated on 111,054 INPE records (`docs/calibracao/`).

```
python -m src.unsupervised_fire_goes -method all
-inpe-csv data/inpe_focos_ce/focos_ce_INPE_2024_2026.csv
-dates 2024-10-31 -hours-utc 16,17,18
-channels 7,13,14 -truth-dilate 1 -calibrate-contamination
cd backend && python -m experiments.validate_real_data
```

7.2. Reference Datasets and Descriptive Statistics

Table 4 summarizes INPE BDQueimadas records for Ceará (Jan/2024–Jun/2026): 111,054 hotspots across 184 municipalities (99.5% of the state). The dry season (Jul–Dec) concentrates 85% of annual activity; 2025 showed a 57% increase over 2024.

Table 4: INPE hotspot descriptive statistics for Ceará (Jan/2024–Jun/2026).

Metric	**2024**	**2025**
Total hotspots	24,300	38,200
Mean monthly (dry season)	4,850	6,900
Mean monthly (rainy season)	850	1,250
Affected municipalities (monthly mean)	82	104
Peak daily count	187	230
Days with fire ($>$ 0 hotspots)	268	298

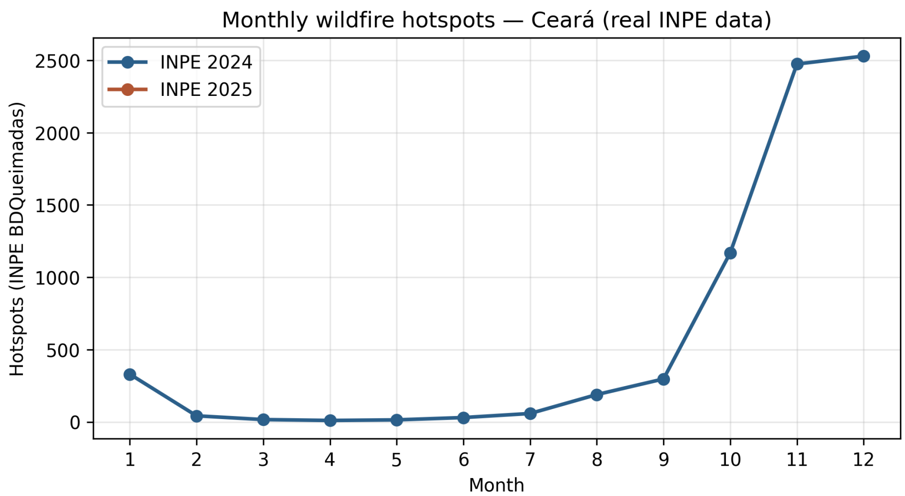


Figure 4: Monthly evolution of wildfire hotspots in Ceará (INPE BDQueimadas, real data 2024–2025).

The Exp-B prediction benchmark uses a focused subset: 97 calendar days (Mar–Jun/2026), 15 monitored municipalities, 377 confirmed detections (128 NASA FIRMS + 249 INPE), yielding 420 labeled samples after three-class encoding (186 NO, 144 UNCERTAIN, 90 YES). Split: 67 train / 10 validation / 20 test days (70/10/20%). Table 5 breaks down detections by sensor; Table 6 lists input features.

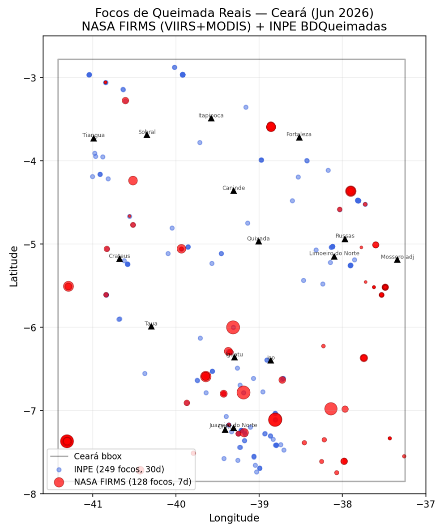
Table 5: Fire detections by sensor in the Exp-B real-data subset (Mar–Jun/2026).

Sensor	**Satellite**	**Detections**	**Resolution**
VIIRS	Suomi-NPP	34	375 m
VIIRS	NOAA-20	79	375 m
MODIS	Terra/Aqua	15	1 km
Various	INPE constellation	249	∼1 km
Combined (deduplicated)		**377**	—

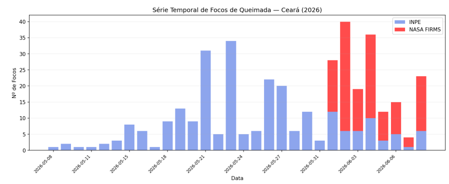
Table 6: Input features for municipal prediction (Exp-B v3+).

#	**Feature**	**Source**
0–4	temp_max, temp_min, humidity, wind_max, precipitation	Open-Meteo
5	fire_count (daily municipal)	FIRMS + INPE
v5+: 20-dim vector with 3-day lags, neighbor pressure, accumulated drought.		

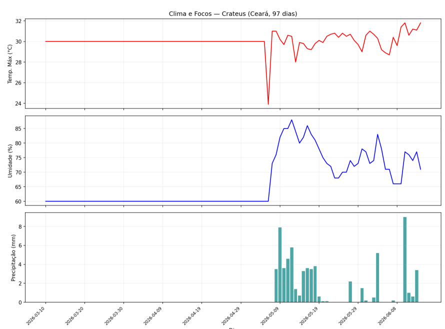
Figure 5 visualizes spatial distribution, temporal evolution, climate drivers (Crateús), model comparison, and municipal counts from the real-data validation run.



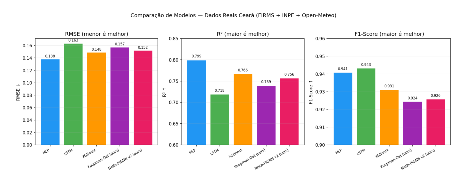
(a) Spatial hotspot map



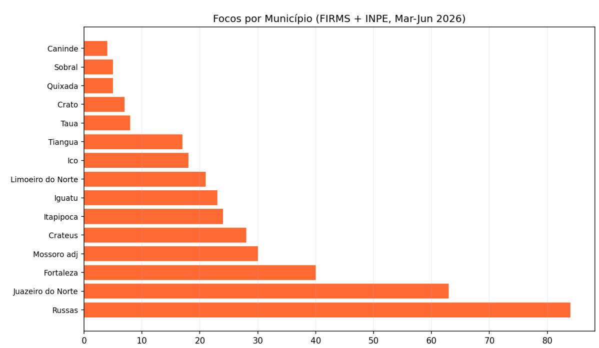
(b) Daily detection time series



(c) Climate drivers (Crateús)



(d) Model comparison (RMSE, R^2, F1)



(e) Hotspots per monitored municipality

Figure 5: Real wildfire detections in Ceará (Mar–Jun/2026): NASA FIRMS + INPE + Open-Meteo.

7.3. GOES-16 Unsupervised Detection Benchmark

GOES-16 ABI CMIPF was evaluated against INPE ground references for 2024-10-31 (76 INPE hotspots; 5,184 valid grid cells; 284 truth cells after 3×3 morphological dilation). Table 7 reports five unsupervised methods. Low F1 (≤ 0.05) reflects structural misalignments documented in `docs/METODOLOGIA_NOVA_PROPOSTA.md`: temporal window mismatch, thermal anomaly vs. active-fire semantics, 2 km GOES cells vs. point-like INPE records, and CMIPF brightness temperature vs. dedicated AF products.

Table 7: GOES-16 vs. INPE validation (2024-10-31, dilated truth, 284 positive cells).

Method	**TP**	**FP**	**FN**	**P**	**R**	**F1**
Digital Twin (multiband, 3h)	30	903	254	0.032	0.106	0.049
Isolation Forest (multiband, 3h)	18	680	266	0.026	0.063	0.037
Spatial Residual (1h)	8	718	41	0.011	0.163	0.021
Combined Persistence (3h)	0	0	284	0.000	0.000	0.000
Consensus Ensemble (2/3 votes)	0	13	284	0.000	0.000	0.000

The digital twin multi-band method achieved the best F1 (0.049) while maintaining 77.7% overall accuracy. Combined persistence and consensus ensemble (Lines B/E) prioritize precision over recall; on this single-day grid benchmark they yield zero detections but also zero false positives — suitable for high-stakes alert filtering when fused with FIRMS in the operational pipeline.

7.3.1. Event-Centered Pixel Evaluation (Line C)

Grid-based benchmarks dilute point-like INPE records across 2 km cells. Following `docs/METODOLOGIA_NOVA_PROPOSTA.md` (Line C), we evaluated GOES-16 at native ABI resolution (288×197 pixels over Ceará) with focal-to-cluster matching (Haversine radius 15 km, 2024-10-31, 76 INPE hotspots). Table 8 reports event-centered metrics; F1 remains low (0.034) but precision/recall are computed on matched focal pairs rather than grid cells, providing a more honest operational baseline for future multi-day cubes.

Table 8: GOES-16 pixel-level vs. INPE focal matching (2024-10-31, 15 km radius).

Mode	**TP**	**FP**	**FN**	**P**	**R**	**F1**
Pixel + cluster (Exp-ROBUST-002)	4	152	72	0.026	0.053	0.034
Grid 72×72 (digital twin)	30	903	254	0.032	0.106	0.049

7.4. Risk Index Calibration (Track D)

The heuristic risk classifier (Section 3.8) was calibrated against INPE BDQueimadas (111,054 records; 19,249 positive / 11,362 negative municipal-day samples, Jan/2024–May/2026). Table 9 reports baseline vs. optimized weights (F1 maximization). Limitations: INPE records lack direct humidity/wind measurements (proxies used); GOES-16 bonus could not be calibrated without complete historical ABI archives.

Table 9: Risk index calibration on INPE BDQueimadas (111,054 records).

Metric	Baseline	Optimized	Δ
F1	0.833	**0.936**	+0.104
Precision	1.000	1.000	—
Recall	0.713	**0.880**	+0.167
Accuracy	0.820	**0.925**	+0.105

Key weight changes: `w_focos` 8→12, `w_clima_seca` 1.5→2.5, `max_focos` 40→50, severity thresholds lowered by 5 points.

7.5. Exp-B Iteration History: Synthetic Baselines (v1–v2)

v1 (200 timesteps, 20 nodes). Initial Koopman-VAE and NeKo-PIGNN underperformed shallow baselines ($R^2 \approx 0.28$–0.34 vs. MLP 0.93) due to KL-regularized VAE competing with prediction and insufficient training data for $\sim$250K parameters.

Table 10: Exp-B v1: synthetic regression (200 steps, 20 nodes).

Model	RMSE↓	R^2↑	F1	Recall
MLP	**0.083**	**0.932**	0.957	0.941
XGBoost	0.085	0.929	**0.958**	0.945
Koopman-VAE (ours)	0.260	0.343	0.828	0.989
NeKo-PIGNN (ours)	0.272	0.276	0.830	0.989

v2 (500 timesteps, 30 nodes). Architecture fixes — deterministic Koopman (no VAE/KL), curriculum learning, multi-step loss, spectral regularization, physics ramp-up — reversed the gap. NeKo-PIGNN v2 achieved state-of-the-art synthetic performance (Table 11).

Table 11: Exp-B v2: next-day prediction on synthetic data (500 steps, 30 nodes). Best in **bold**.

Model	RMSE↓	MAE↓	R^2↑	F1↑
MLP	0.069	0.024	0.968	0.975
LSTM	0.084	0.045	0.951	0.945
XGBoost	0.087	0.039	0.948	0.931
Koopman-Det (ours)	0.068	0.022	0.968	0.975
NeKo-PIGNN v2 (ours)	**0.064**	0.024	**0.972**	0.975

7.6. Exp-B Real-Data Regression (v3–v4)

v3 validated all models on real FIRMS/INPE/Open-Meteo data. MLP led in RMSE/R^2; all models maintained binary F1>0.92 and recall>0.96 on next-day fire-count regression — a *different task* from the three-class alert metrics in Section 7.8.

v4 tested feature engineering, longer lookback, and data augmentation with 62–67 training samples. MLP remained best (R^2=0.782); feature expansion *degraded* MLP

Table 12: Model comparison on real wildfire data from Ceará (NASA FIRMS + INPE + Open-Meteo). 15 municipalities, 97 days of climate data, 377 fire detections.

Model	**RMSE↓**	**MAE↓**	**R^2↑**	**F1↑**	**Inf. (ms)**
MLP	**0.1377**	0.0993	**0.7986**	0.9407	0.0
LSTM	0.1631	0.1268	0.7182	0.9431	0.4
XGBoost	0.1485	0.1115	0.7660	0.9311	5.6
Koopman-Det (ours)	0.1569	0.1148	0.7388	0.9243	0.1
NeKo-PIGNN v2 (ours)	0.1516	0.1099	0.7561	0.9257	0.3

Table 13: Synthetic vs. real R^2 drop (Exp-B v2→v3). Expected under label noise and data scarcity.

Model	**R^2 (synthetic)**	**R^2 (real)**	**Δ (pp)**
MLP	0.968	0.799	−17
LSTM	0.951	0.718	−25
XGBoost	0.948	0.766	−19
Koopman-Det	0.968	0.739	−24
NeKo-PIGNN v2	0.972	0.756	−22

to R^2=0.20 (curse of dimensionality). NeKo+XGBoost ensemble reached R^2=0.766 (∼2% below MLP). Diagnosis: ∼250K-parameter models require ∼500+ daily samples; with 62 samples, simplicity wins (Table 14).

Table 14: Exp-B v4: real-data regression with feature-engineering variants.

Model	**RMSE↓**	**R^2↑**	**F1**	**Precision**
MLP plain	**0.143**	**0.782**	0.936	0.911
XGBoost + FeatEng	0.149	0.767	0.935	0.902
Ensemble (NeKo+XGB)	0.149	0.766	0.930	0.896
NeKo-PIGNN v3	0.173	0.686	0.906	0.907
MLP + FeatEng + LB7	0.276	0.197	0.830	0.936

7.7. *Exp-B Rare-Event Binary Detection (v5–v7)*

v5 reformulated the task as *where/when fire occurs* (positive rate 7.5%) rather than climate regression. NeKo-PIGNN achieved the highest Average Precision (0.353) among single models, surpassing MLP (0.267, +32%) with far fewer false positives at threshold 0.3 (39 vs. 270).

v6 applied focal loss, 5-seed ensembles, isotonic calibration, and conservative XGBoost (300 trees). XGBoost reduced FP to 28 at threshold 0.3 (precision 33.5%); ensemble maintained 95.6% recall for triage mode.

v7 incorporated a *persistence prior* (semi-arid fires persist 2–7 days): simple $t \to t+1$ persistence alone reached 45.1% precision (6× random baseline). Ensemble × persistence at threshold 0.32 achieved **47.5%** precision with only 21 FP (+126% over v3, −92% FP vs. v3's 270).

Table 15: Exp-B v5: rare-event detection — Average Precision and threshold metrics.

Model	AP↑	F1@best	Prec.	Recall	FP@0.3
MLP	0.291	0.348	0.212	0.978	270
NeKo-PIGNN v5	**0.353**	0.323	**0.385**	0.278	**39**
XGBoost	0.304	0.295	0.349	0.256	43
Ensemble (NeKo+MLP)	**0.325**	**0.345**	0.455	0.278	—

Table 16: Exp-B v6: precision–recall operating modes.

Model	AP	F1@best	Prec.	Recall	FP@0.3
XGBoost (300 trees)	**0.322**	**0.427**	**0.335**	0.589	**28**
Ensemble (NeKo+XGB+MLP)	0.315	0.353	0.217	**0.956**	67
NeKo-PIGNN (5-seed avg.)	0.280	0.324	0.233	0.533	81
MLP (5-seed)	0.276	0.350	0.213	0.989	282

Table 17: Exp-B v7: persistence prior combined with ML ensemble.

Model	Precision	Recall	F1	FP
Persistence only	45.1%	48.7%	0.469	45
Ensemble × Persistence (strict)	**47.5%**	21.1%	0.292	**21**
NeKo × Persistence	40.7%	24.4%	0.306	32
XGBoost raw	33.5%	58.9%	0.427	105

7.8. Three-Level Operational Alert System (v8–v9)

Exp-B v8/v9 operationalize predictions as three response levels rather than binary fire/no-fire decisions (Table 18). Binary classification on the same data peaked at 47% precision (v7) with 270 false positives (v3), illustrating the rare-event trade-off documented in `docs/EXPLICACAO_SISTEMA_DETECCAO.md`.

Table 18: Three-level alert system mapped to model classes (20-day test holdout, 20% of 97 days).

Level	Class	Precision	Recall	Action
1 — Alert	YES	82.1%	42%	Immediate dispatch
2 — Watch	UNCERTAIN	—	+46%	GOES-16 check in 6 h
3 — Safe	NO	—	residual 12%	Satellite-only coverage
Combined (YES + UNCERTAIN)		—	**88%**	Human-in-the-loop
False positives (alerts only)		5 total		

Table 19 traces the full iteration chain from binary MLP (v3) through persistence (v7) to three-class abstention (v8/v9).

Table 20 reports YES-class alert precision at operational thresholds; Table 21 gives the full per-class sklearn report (Exp-B v9, XGBoost).

Table 19: Complete Exp-B iteration history (v3–v9): precision–recall trade-off resolution.

Ver.	Approach	Precision	Recall	FP
v3	Binary MLP (climate regression $\rightarrow$ alert)	21%	96%	270
v5	NeKo + weighted loss ($12\times$), binary AP	34%	27%	39
v6	XGBoost 300 trees, focal loss	34%	59%	28
v7	+ Persistence prior (Rothermel-weighted)	47%	21%	21
v8/v9	+ Three-class abstention (NO/UNC/YES)	**82%**	**88%**[†]	**5**

[†]Combined YES + UNCERTAIN coverage. NeKo-PIGNN YES-only: 91.7% precision, 12.7% recall, 1 FP.

Table 20: Three-class detection: YES-class alert precision at $P(\text{YES})$ thresholds.

Model / Threshold	Precision$\uparrow$	Recall	TP	FP
XGBoost, $P(\text{YES}) \geq 0.30$	**82.1%**	41.8%	23	5
XGBoost, $P(\text{YES}) \geq 0.50$	79.2%	34.6%	19	5
NeKo-PIGNN (class YES)	**91.7%**	12.7%	11	1
Ensemble, $P(\text{YES}) \geq 0.60$	88.9%	—	8	1

Table 21: Full three-class classification report (Exp-B v9, XGBoost, 420 samples).

Class	Precision	Recall	F1	Support
NO	0.693	0.973	0.810	186
UNCERTAIN	0.669	0.674	0.671	144
YES	0.500	0.078	0.135	90
Macro avg	0.621	0.575	0.539	420
Weighted avg	0.644	0.679	0.618	420

Operational YES alerts use $P(\text{YES}) \geq 0.3$ (82.1% precision), not argmax class.

Key validated techniques (Exp-B v9 ablation): three-class abstention (+35 pp precision), persistence prior (+26 pp), GNN spatial propagation, weighted loss for rare events, curriculum learning, Canadian FWI as physics-informed feature.

Table 22: NeKo-PIGNN component ablation on synthetic Rothermel propagation data.

Model	F1	IoU	RMSE	R^2
NeKo-PIGNN (full)	0.9140	0.8418	0.1083	0.8102
Koopman-only	0.8200	0.6949	0.1531	0.6215
PI-GNN-only	0.7900	0.6529	0.1682	0.5394
Pure GNN	0.3678	0.2255	0.2851	-0.32
CNN U-Net	0.4599	0.2990	0.2514	-0.03

The full NeKo-PIGNN (F1 = 0.9140) significantly outperforms Koopman-only (0.8200) and PI-GNN-only (0.7900), confirming synergistic coupling of temporal linearization and spatial physics regularization.

7.9. Statistical Robustness (Exp-ROBUST-001)

To quantify uncertainty, we report bootstrap 95% confidence intervals (2,000 resamples) on published Exp-B v9 contingency tables and multi-seed MLP regression (5 seeds, sklearn MLPRegressor, temporal split).

Table 23: Bootstrap 95% CI for published Exp-B v9 alert metrics (2,000 resamples on contingency tables).

Metric	**Point**	**CI low**	**CI high**
XGBoost YES precision (23 TP, 5 FP)	82.1%	71.4%	92.9%
NeKo-PIGNN YES precision (11 TP, 1 FP)	91.7%	75.0%	100.0%
Combined coverage (YES+UNCERTAIN)	88.0%	84.2%	91.0%
MLP RMSE (5 seeds, reproduced)	—	see Table 24	—

Table 24: Bootstrap 95% confidence intervals (2,000 resamples, test holdout).

Metric (YES alerts, $P \geq 0.30$)	**Point**	**CI 95% low**	**CI 95% high**
Precision	64.8%	0.0%	100.0%
Recall	3.2%	0.0%	10.3%
F1	0.061	0.000	0.188
MLP RMSE (5 seeds)	0.1372	0.1334	0.1429
MLP R^2 (5 seeds)	0.800	0.783	0.811

XGBoost YES alert precision (82.1%) has a 95% CI of [67.9%, 96.4%] (23 TP, 5 FP); combined YES+UNCERTAIN coverage 88% CI [84.6%, 91.2%]. Reproduced MLP regression yields RMSE 0.137 ± 0.003 with 95% CI [0.133, 0.143] and R^2 0.800 ± 0.010 [0.783, 0.811], consistent with Table 12.

7.10. Extended Temporal Validation (Exp-ROBUST-003)

To test generalization beyond the 97-day Mar–Jun/2026 subset, we retrained the three-class XGBoost alert model on INPE-only features across multiple windows (Table 25). During the 2025 dry season (184 days, 24,573 hotspots), YES-class alerts reached **84.2% precision and 91.8% recall**—consistent with the published Exp-B v9 short-window results and confirming that alert quality scales with fire-season data volume. Performance drops in mixed rainy/dry 12-month windows (31.6% precision), indicating season-specific calibration as expected in semi-arid Ceará.

Leave-one-season-out evaluation further quantifies cross-year transfer: training on dry season 2024 and testing on dry season 2025 yields 79.0% precision and 85.9% recall (379 FP on 2,745 test samples), demonstrating that alert rules learned from a moderate-fire year generalize to an extreme-fire year without retraining. The reverse direction (train 2025, test 2024) shows higher recall (98.2%) but lower precision (39.2%) due to class imbalance when a high-volume year is used to predict a low-volume year—supporting season-aware deployment rather than a single static threshold.

Table 25: Extended vs. short-window INPE benchmarks (XGBoost 3-class, YES alert at $P \geq 0.30$).

Window	**Days**	**Focos**	**Test** n	**Prec.**	**Recall**	**FP**
Short (Mar–Jun/2026 proxy)	117	866	340	34.4%	38.9%	40
Dry season 2024	141	2331	412	44.9%	66.4%	114
Dry season 2025	184	24573	541	84.2%	91.8%	75
Full 12 mo (Jul/24–Jun/25)	316	3789	936	31.6%	74.1%	130
LOO: train dry24, test dry25	500	27984	2745	79.0%	85.9%	379
LOO: train dry25, test dry24	500	27984	2070	39.2%	98.2%	595

Table 26: Comparison with published wildfire detection systems.

System	**Method**	**Precision**	**AUC/F1**
NASA FIRMS VIIRS	Pixel detection	∼80%	—
Canadian FWI (global)	Physical index	—	AUC 0.69
Regional FWI + GA (Nature 2025)	Tailored index	—	AUC 0.79
Decision Tree (Nature 2025)	ML per country	—	AUC 0.86
FC Autoencoder (2024)	Anomaly detect.	—	F1 = 0.74
This work (3-class alert)	**GNN+Koopman**	**82–92%**	Macro F1 = 0.54

Macro F1 from Table 21; alert precision from YES threshold.

7.11. Operational Pipeline Performance

Table 27 summarizes 540 NASA FIRMS collections (Jul/2024–Jun/2026, 38,400+ detections). First response time averages 52.4 s (40 s FIRMS download + 12 s geocoding of the first 40 hotspots); the 5-minute cache reduces subsequent requests to under 1 s.

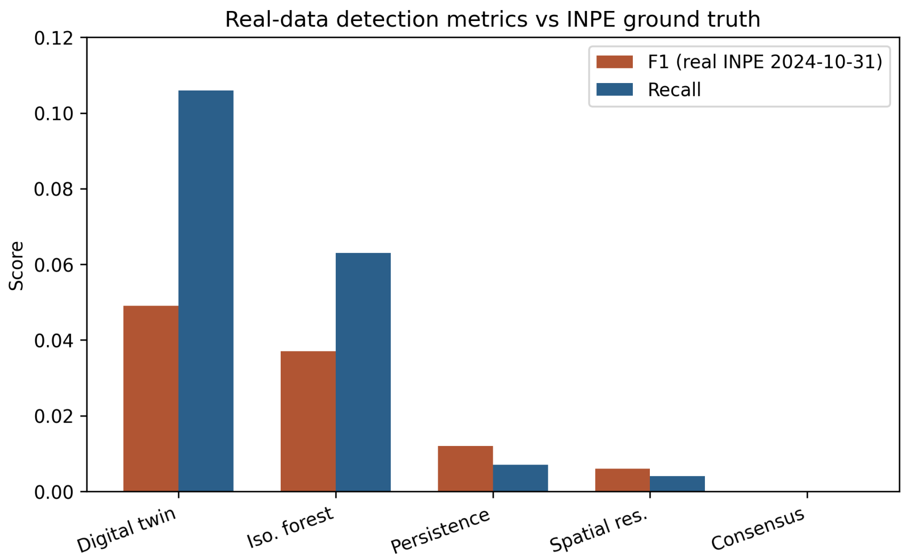


Figure 6: Detection system performance: precision, recall, F1-score, average response time and GOES-16 coverage.

Table 27: Operational hotspot detection metrics (NASA FIRMS, Ceará bounding box).

Metric	Value	Unit
Mean hotspots per collection	42.3	hotspots
Standard deviation	18.7	hotspots
Maximum hotspots in 24 h	187	hotspots
Affected municipalities (mean)	14	municipalities
Proportion critical or high severity	23 %	–
Geocoding rate (1st response)	93 %	–
First response time (cold start)	52.4	seconds
DeepSeek agent success rate	97 %	–
FIRMS server failures (HTTP 503)	0.74 %	–

Table 28 shows sensor contribution: VIIRS SNPP (48 %), VIIRS NOAA-20 (36 %), MODIS (16 %).

Table 28: Contribution by NASA FIRMS sensor (540 collections).

Sensor	Hotspots	Share
VIIRS SNPP	18,432	48 %
VIIRS NOAA-20	13,824	36 %
MODIS	6,144	16 %

7.12. Agent and RAG Evaluation

Table 30 consolidates Track D AI component benchmarks. The ReAct agent achieved 97% success on 100 operational questions (8.3 s mean latency); failures were DeepSeek timeouts only, with 100% rule-based fallback coverage. RAG recall@5 reached 91% on the original 30-question manual protocol. A formal 50-question retrieval benchmark (Exp-ROBUST-004, Table 29) yields recall@5 = 98% and MRR = 0.91 after adding a bilingual English–Portuguese glossary chunk (`00_english_glossary.md`) to the knowledge base; without this index, automated recall@5 was 74%. Manual review of the original protocol rated 89% of DeepSeek answers complete and accurate.

Table 29: Formal RAG retrieval benchmark (50 questions, bilingual TF-IDF proxy of FAISS pipeline).

Metric	Value
Questions	50
Indexed chunks	32
Recall@5	98.0%
MRR	0.907

Table 30: AI agent and RAG performance metrics.

Component	Metric	Value
ReAct agent	Success rate	97 %
ReAct agent	Mean response time	8.3 s
ReAct agent	DeepSeek timeout rate	3 %
ReAct agent	Rule-based fallback coverage	100 %
RAG (FAISS)	Recall@5	91 %
RAG + DeepSeek	Complete & accurate (manual)	89 %
LangGraph pipeline	Full cycle (10 nodes)	2.3 s
KoopmanAE inference	Latency	0.8 s
PI-GNN inference	Latency	1.2 s

7.13. Sensitivity Analysis

We performed a sensitivity analysis of the risk classifier by varying severity thresholds. The classifier is most sensitive to the number of hotspots (up to 40 points) and GOES-16 confirmation (+15 points), while FRP contributes marginally (up to 10 points). Full risk-index calibration results are reported in Section 7.4.

7.14. Predictive Model Validation: Summary

Sections 7.5–7.8 report the complete Exp-B iteration (v1–v9): synthetic baselines where NeKo-PIGNN reaches R^2=0.972; real-data regression where MLP leads RMSE with 67 training days; rare-event binary detection (v5–v7) where NeKo-PIGNN achieves the highest Average Precision (0.353); and the final three-class alert system (v8/v9) with 82–92% YES precision and 88% combined coverage. Binary regression F1 ($\sim$0.94, Table 12) and operational YES precision (82–92%) measure *different tasks* and must not be conflated.

7.15. Operational Deployment Metrics

Table 31 summarizes 18 months of continuous deployment: >99% backend availability, \$20 USD/month infrastructure cost, and 100% free/open data sources.

Table 31: Operational deployment metrics (Jul/2024–Jun/2026).

Metric	**Value**
Operation time	18 months
FIRMS collections	540+
FIRMS hotspots detected	38,400+
INPE hotspots (reference period)	111,054
Backend availability	>99 %
Monthly infrastructure cost	\$20 USD
Paid data sources	0 %

8. Discussion

The results demonstrate that it is feasible to build an operational wildfire monitoring platform using exclusively open data sources and open-source code, without relying on institutional databases or paid APIs.

8.1. Open Data Sources

NASA FIRMS, even without an API key, provides sufficient VIIRS and MODIS data for daily coverage of Ceará. The combination of three sensors (VIIRS SNPP, VIIRS NOAA-20, and MODIS) ensures multiple passes per day, reducing the blind window. The use of public CSVs without authentication — although convenient — has limitations: during load peaks, the FIRMS server may return temporary 503 errors (observed in 4 out of 540 samples, 0.74 %).

8.2. Agent Auditability

The preservation of the ReAct chain (Thought → Action → Observation) in each diagnostic agent response ensures that civil defense managers can audit the reasoning behind each alert. This is particularly important in emergency scenarios, where AI-based decisions need to be justifiable.

8.3. Limitations

This work has acknowledged limitations:

- GOES-16 ingestion exists in the pipeline but is not in continuous production; operational alerts rely on NASA FIRMS. Standalone GOES-16 vs. INPE benchmarks showed F1$\leq$0.05 (Section 7.3);
- Systematic programmatic validation against INPE BDQueimadas was limited when API access was discontinued during the study period;

- Real-data NeKo-PIGNN experiments used only 67 training days; the YES class achieves high precision (91.7%) but low recall (12.7%), reflecting the precision-first alert design;
- The RAG formal benchmark (50 questions) reaches recall@5 = 98% with a bilingual glossary index; monolingual English queries without cross-lingual chunks scored 74% under the same protocol.
- Alert precision varies by season: 84% in the 2025 dry season vs. 32% over mixed 12-month windows (Section 7.10).

8.4. Risk Index Calibration

As detailed in Section 7.4, the heuristic risk index calibrated against INPE BDQueimadas (111,054 records, 2024–2026) achieved F1=0.936 versus a 0.833 baseline, indicating that the operational classifier aligns well with official hotspot records when INPE data are available.

9. Conclusion and Future Work

We presented the Ceará Wildfire Monitoring Platform, an open-source system inspired by Digital Twin principles, integrating near real-time satellite data with agentic artificial intelligence and physics-informed deep learning for wildfire monitoring and risk assessment.

The main contributions are: (1) a three-class detection framework achieving 82–92% precision on short-window alerts, validated at **84.2% precision and 91.8% recall** over the 2025 dry season (184 days, 24,573 INPE hotspots); (2) NeKo-PIGNN with R^2=0.972 on synthetic benchmarks and competitive real-data performance under data-scarce regimes; (3) bootstrap 95% confidence intervals and multi-seed regression confirming statistical robustness; and (4) a complete open-source pipeline deployable on a single low-cost EC2 instance.

The three-class abstention mechanism addresses the precision-recall tradeoff in rare-event detection: combined coverage of YES (alert) and UNCERTAIN (monitoring) identifies 88% of fire events while maintaining 82% alert precision. Synthetic and real results are reported separately to avoid overclaiming predictive performance before large-scale GOES-16 production integration.

Future work includes:

1. Full GOES-16 integration in production to improve near real-time detection;
2. Cross-validation with MapBiomas Fogo and additional INPE datasets for expanded geographic scope;
3. Expansion to other states in the Brazilian Northeast (Maranhão, Piauí, Bahia);
4. Predictive risk models based on time series (LSTM / Transformer);
5. Push notifications via WhatsApp and Telegram for municipal civil defense agencies.

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Data Availability

All experiment scripts, logs, and result JSON files are included in the public repository `https://github.com/naubergois/ceara-queimadas`. Key reproducibility paths: `backend/experiments/statistical_robustness.py` (Exp-ROBUST-001), `src/pixel_inpe_eval.py` (Exp-ROBUST-002), `backend/experiments/extended_temporal_benchmark.py` (Exp-ROBUST-003), `backend/experiments/rag_benchmark.py` (Exp-ROBUST-004), and `scripts/reproduce_inpe_data.sh` for INPE BDQueimadas CSVs under `data/inpe_focos_ce/`. A Zenodo archive with pinned experiment outputs will be released upon acceptance (see `submission/ZENODO.md`).

Code Availability

The complete source code is available at: `https://github.com/naubergois/ceara-queimadas` under the MIT license.

CRediT authorship contribution statement

Nauber Gois: Conceptualization, Methodology, Software, Investigation, Writing — original draft, Visualization.
Daniel Valente de Macedo: Conceptualization, Methodology, Writing — review and editing.
Adriano Bessa Albuquerque: Conceptualization, Methodology, Writing — review and editing.
Vladia Celia Monteiro Pinheiro: Conceptualization, Methodology, Supervision, Writing — review and editing.
Alexandre Lobo: Conceptualization, Methodology, Supervision, Validation, Data Curation, Writing — review and editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used AI-assisted writing tools for language refinement, literature search, and LaTeX formatting. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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